High-fidelity libraries for enzyme optimization

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Abstract

Science abstract target: ∼125 words. Structure aligned to the three-application spine. Written last, on purpose — it should compress the actual results rather than lead them.

- **Hook.** De novo enzyme design produces backbones; optimization is the bottleneck that prevents most from reaching useful activity.
- **Approach.** Function-aligned RL fine-tuning of ProteinMPNN, combined with fragment-based combinatorial libraries and a closed-loop experimental optimization protocol.
- **Result (a) — heme oxidation.** Fine-tuned MPNN outperforms SSM and vanilla MPNN at matched experimental effort.
- **Result (b) — PETase.** Experimental data from an initial library trains a surrogate that drives a second library to substantially better designs (XXX-fold over starting; exceeds optimized natural enzymes on the PET-trimer probe).
- **Result (c) — protease.** A batch-consensus mutation-diversity reward unlocks rarer mutations and improves catalytic activity over an equivalent library without the diversity term.
- **Implication.** Broad, function-aligned sequence-space search plus an experimental closed loop is a general strategy for de novo enzymes.

[TODO: abstract — fill once results sections converge]

Contents

Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	**drafted**
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	**drafted**
`Makefile`	`make pdf`; SVG→PDF via Inkscape	**drafted**
`refs.bib`	Bibliography; all entries machine-retrieved	**drafted**
Front matter		
`abstract.tex`	∼125-word abstract, written last	**scaffold**
`intro.tex`	Four-paragraph introduction arc	**scaffold**
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	**drafted**
Results — compile order is risk order		
`results/ame.tex`	AME benchmark; diversity at matched fidelity; fully computational and reproducible	**scaffold**
`results/heme.tex`	Three-arm head-to-head; foundational claim	**needs data**
`results/petase.tex`	Closed-loop iteration; strongest data	**scaffold**
`results/protease.tex`	Diversity-reward ablation; highest risk	**needs data**
Methods		
`methods/overview.tex`	GRPO, fragment libraries, surrogate, diversity reward	**scaffold**
`methods/heme.tex`	Design config, three-arm protocol, assay	**scaffold**
`methods/petase.tex`	Two libraries, surrogate, assays	**scaffold**
`methods/protease.tex`	Design config, two-arm ablation, assay	**scaffold**
`experiments.tex`	Arms, selection algorithms, experiment slate E1–E14 with status	**scaffold**
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	**scaffold**

Two companion files live in `paper/` but are deliberately *not* part of this build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

Science research articles do not use `\section` labels in the main narrative. The headings here exist so that this draft has a usable table of contents; strip them at submission time. Paragraph-level scaffolding is kept visible so collaborators can see the intended arc.

P1 — The promise and the bottleneck

- RFdiffusion + MPNN make it possible to design enzyme backbones from scratch for diverse chemistries.
- First-pass designs are typically weak; the bottleneck is optimization, not initial design.

[TODO: p1 — frame de novo enzyme design + the optimization bottleneck]

P2 — Why existing optimization tools fall short

- Site-saturation mutagenesis (SSM): local; cannot escape the neighborhood of the starting design.
- Vanilla MPNN re-sampling: structurally informed but functionally blind; biased toward sequences MPNN's prior favors, not sequences that catalyze.
- The active-variant region for a de novo backbone often lies tens of mutations away — out of reach of either tool.
- Cross-reference Section 2 for the full treatment; this paragraph is the compressed version.

[TODO: p2 — limits of SSM and vanilla MPNN re-sampling]

P3 — What we contribute (three orthogonal capabilities)

- Function-aligned RL fine-tuning of ProteinMPNN (GRPO over reward pipelines) + fragment-based combinatorial libraries that can physically test combinatorially large candidate sets.
- **(i) Method foundation** — heme oxidation: fine-tuned MPNN beats SSM and vanilla MPNN at matched experimental effort.
- **(ii) Closed-loop iteration** — PETase: an initial experimental library trains a surrogate that becomes a reward signal for the next library, producing substantially improved designs.
- **(iii) Diversity-aware search** — protease: a batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library.

[TODO: p3 — three contributions, framed as separate claims]

P4 — Headline numbers + roadmap

- Heme: ⟨delta over SSM / vanilla MPNN at matched library size⟩.
- PETase: XXX-fold k_{cat} improvement on PET-trimer probe; best variant $\sim$30 mutations from starting; exceeds optimized natural enzymes on the same substrate.
- Protease: ⟨delta in activity for diversity-reward arm vs. baseline arm⟩.
- Together: a generalizable recipe for de novo enzyme optimization.

[TODO: p4 — headline results across the three applications]

2 Status quo: where the field stands and where the gap is

This section is written as real prose (not scaffold) because it is the part of the paper that does not depend on our own unfinished data. Every citation here was verified against Crossref, the arXiv API, or Europe PMC on 2026-08-04; the retrieval procedure is documented at the top of `refs.bib`. Quantitative claims attributed to other groups are taken verbatim from their abstracts.

2.1 Backbone generation is solved well enough that sequence design is now the bottleneck

Deep generative models now produce enzyme backbones for a wide range of chemistries. RFdiffusion established de novo design of protein structure and function by denoising diffusion over backbone coordinates [1]. RFdiffusion2 removed the requirement that catalytic geometry be specified at the residue level: it designs enzymes directly from functional-group geometries without specifying residue order or performing inverse rotamer generation, and generates scaffolds for all 41 active sites in a diverse in-silico benchmark, compared with 16 for previous methods [2]. Parallel efforts have produced designed serine hydrolases [3], metallohydrolases with designed metal sites [4], and enzymes built by catalytic motif scaffolding [5].

> The rhetorical work of this subsection is to concede generously. These are strong papers and several share our senior author. The point is not that backbone generation is inadequate — it is that its success relocates the bottleneck downstream, to sequence design and to what happens after the first 96 designs come back weakly active.

Two features of this literature matter for what follows. First, the sequence-design step is treated as a solved subroutine: backbones are passed to ProteinMPNN [6] or LigandMPNN [7] sampled at low temperature, and the resulting sequences are filtered against structure-prediction and geometric criteria. Second, the experimental readout is a small, one-shot test set: RFdiffusion2 reports active candidates after testing fewer than 96 sequences per catalytic mechanism [2]. Both features are appropriate for demonstrating that a backbone generator works. Neither addresses what to do when the resulting enzymes are, as they almost always are, orders of magnitude less active than useful.

2.2 Filtering is not search

The prevailing sequence-design protocol is generate-then-filter. Sequences are sampled from a functionally blind inverse-folding prior, then scored against structure-prediction confidence, backbone recapitulation, catalytic-geometry and preorganization criteria — AlphaFold3 [8], Boltz [9], and ensemble methods such as PLACER, whose introduction substantially improved experimental success rates in serine hydrolase design [3].

> This is the paper's central rhetorical move, so state it plainly and early, and make sure the phrasing survives review: a filter is a selection operator, not a search operator.

A filter cannot create a sequence that passes; it can only surface the sequences that a functionally blind prior happened to emit. This has two consequences that the field has largely left implicit. Sampling temperature is lowered to raise the pass rate, which means pass rate is purchased directly with diversity: ProteinMPNN's own report notes that sequence diversity can be increased considerably at higher temperature with only a small decrease in recovery [6], and the corresponding pass-rate cost is why production pipelines sit near $T = 0.1$. And because the survivors are drawn from a narrow neighborhood of the prior's mode, they are strongly correlated with one another — so a library's nominal size overstates the number of genuinely independent attempts it represents.

2.3 Reinforcement learning has reached inverse folding, but not libraries

Aligning generative models to objectives via reinforcement learning is now standard practice, and Group Relative Policy Optimization (GRPO) [10] in particular has proven to be a memory-efficient fit for the regime where rewards are computed by an external oracle rather than learned. This machinery has begun to reach protein sequence design. ProteinZero is an online RL framework for inverse folding that combines structural guidance from ESMFold with a self-derived $\Delta\Delta G$ predictor, adds an embedding-level diversity regularizer to mitigate mode collapse, and reports reducing design failure rates by 36–48% relative to ProteinMPNN, ESM-IF [11], and InstructPLM on the CATH-4.3 benchmark, with success rates above 90% across diverse folds [12]. Diversity-regularized direct preference optimization has been applied to peptide inverse folding with related motivation [13].

ProteinZero is the closest prior art to our mechanism and must be confronted in the first paragraph a reviewer reads on this topic, not buried. The differentiation is real but it is about the *objective*, not the optimizer: (i) general folds on CATH vs. enzyme active sites with ligands and catalytic geometry; (ii) sequence-level metrics — designability, recovery, stability — vs. a library-level objective; (iii) a diversity *regularizer* defended as anti-mode-collapse hygiene vs. diversity as a first-class design goal justified by screening economics; (iv) no physical library. If compute allows, running ProteinZero as a baseline arm is the strongest possible version of this argument. **[TODO: decide whether ProteinZero is runnable as a baseline]**

What this line of work does not address is the library. Its objectives and benchmarks are defined per sequence: is *this* sequence designable, stable, recovered. The quantity that determines an experimental outcome is a property of the set — how many distinct, genuinely different candidates clear the bar, and how far apart they are.

2.4 Library design assumes a natural enzyme

A mature literature does optimize libraries, but from the opposite starting point. MODIFY learns from natural protein sequences to infer evolutionarily plausible mutations and predict enzyme fitness, and explicitly co-optimizes predicted fitness and sequence diversity of starting libraries, prioritizing high-fitness variants while ensuring broad sequence coverage [14]. Active Learning-assisted Directed Evolution (ALDE) uses uncertainty quantification over an epistatic five-residue active site and improved the yield of a non-native cyclopropanation product from 12% to 93% in three rounds of wet-lab experimentation [15]. Machine-learning-guided cell-free expression platforms scale the assay side, mapping fitness across large variant sets [16].

MODIFY is the single most important citation for our diversity claim, because it proves the field already accepts that fitness and diversity should be co-optimized in a library. That is helpful — we are not arguing for diversity from scratch. What we are arguing is that every existing method for doing so is unavailable in the de novo regime.

Every one of these methods presupposes what a de novo enzyme lacks. MODIFY's fitness prior is evolutionary, learned from natural homologs; a de novo backbone has no homologs and no alignment. ALDE and active-learning methods require a measurable starting fitness to bootstrap from; de novo designs typically start at or near the detection limit. And both operate over a handful of positions, whereas the mutations that rescue a de novo enzyme are, in our hands, tens of substitutions from the starting design — a combinatorial volume that site-saturation and few-position libraries cannot reach.

2.5 The gap

Nobody has addressed library design for de novo enzymes. The regime is defined by three simultaneous absences: no evolutionary prior, so MSA-based fitness models do not apply; no starting activity, so active-learning loops have nothing to condition on; and a target region of sequence space too far away for local mutagenesis. The field's current answer — low-temperature inverse folding plus hard filters — is not a search procedure at all, and it responds to the difficulty of the regime by narrowing the search precisely when it should be widening it.

Close on the thesis so the reader carries it into the results. Draft formulation, to be tightened once F2 exists: function-aligned online RL converts in-silico filters from a rejection step into a design objective, producing de novo enzyme libraries that simultaneously pass the community's own fidelity criteria at higher rates and span far more sequence space — increasing the number of independent experimental bets per unit of screening effort.

[TODO: gap paragraph — final phrasing of the thesis sentence, once the headline figure is built]

3 Results: the AME benchmark (diversity at matched fidelity)

Claim. On an established public benchmark, CLEO produces libraries with the sequence coverage of high-temperature LigandMPNN and the pass rate of low-temperature LigandMPNN — a combination no single temperature achieves. **Why this section exists.** The three application sections rest on experimental assays that take months and cannot be re-run by a referee. This one is fully computational, uses a published benchmark and a published deposit, and can be reproduced end to end. It is the section that makes the method credible before any wet-lab claim is evaluated. Status: scaffold — the CLEO arm is measured, the temperature sweep is not yet run.

Block 1 — Setup and the unit of comparison

We evaluate on the atomized motif enforcement (AME) benchmark introduced with RFdiffusion2, comprising 41 catalytic sites scaffolded into 100 backbones each. The published protocol assigns 8 LigandMPNN sequences per backbone and predicts each with 5 Chai models; a prediction succeeds when the motif all-atom RMSD to the design is below 1.5 Å and no backbone atom clashes with the ligand.

The aggregation rule matters more than it looks and is easy to get wrong. We verified it against the deposit rather than assuming: the per-site number reported in `AME_benchmark_summary.csv` is the fraction of the 100 backbones with *at least one* passing prediction out of the 40, matching to zero error across all 41 sites and both methods. It is a best-of-40, not a mean over predictions. Scoring one prediction per sequence and comparing rates therefore understates a comparable method by roughly threefold — in the deposit, only 36 % of the sequences that pass on some Chai model pass on all five. Every number below is on their unit.

- Pilot backbones: three drawn from the `pass` class, at 32/40, 28/40 and 10/40 passing predictions in the deposit.
- Our predictor is AlphaFold3 rather than Chai; best-of-N uses 5 diffusion samples per sequence.
- **Caveat to resolve before submission.** RFdiffusion2 scores against the unidealized, packed design; we score against the idealized backbone. This is the leading explanation for two of the three pilot backbones failing in our pipeline at the published temperature despite passing in the deposit. Until it is closed, only the backbone that reproduces (M0097) supports a head-to-head claim.

[TODO: ame — close the idealized/unidealized reference-structure discrepancy]

Block 2 — Low-temperature sampling returns one solution, sampled repeatedly

The incumbent protocol samples LigandMPNN at low temperature. At $n = 96$ on M0097 this yields a high success rate — 58.3 % of sequences pass — but the passing designs are a tight cluster: mean pairwise Hamming distance 40.6 out of $L = 180$, and 162 distinct substitutions among any 15 of them. The library is close to one solution measured many times, and most of the folding budget spent distinguishing its members buys little.

Raising the temperature removes the redundancy and most of the signal with it. At $T = 1.0$ the same backbone reaches mean pairwise distance 126.3 at a success rate of 4.2 %. Diversity alone is not the objective; a library that cannot be filtered is not a library.

Two numbers here were wrong in earlier drafts and are corrected. $T = 1.0$ was reported as *zero* passing designs; that came from $n = 8$, and at $n = 96$ it passes 4.2 % of the time. And the low-temperature arm was described as collapsing to a single cluster "at every threshold down to 50 % identity" — true at 70 %, but at 90 % every passing design is its own cluster and the statistic simply counts designs. Cluster counts are threshold-dependent and are no longer quoted; mean pairwise Hamming and the distinct-substitution count U are threshold-free and replace them throughout.

Block 3 — cleo occupies the corner neither temperature reaches

Sampling temperature traces a frontier: success falls as diversity rises, and every point on it is available to the incumbent by turning one knob. The claim of this section is that CLEO sits above that frontier, and the measurement is in Table 1.

Table 1: M0097, per-sequence success rate on best-of-5 AF3 predictions, with diversity measured over the *passing* designs only — a library contains only what survives filtering. U_{15} is the number of distinct (position, residue) substitutions in a library of 15 passing designs, rarefied over 300 draws so arms with different yields are comparable. Baselines are $n = 96$; CLEO arms are the final 40 steps of a 200-step run.

Arm	**Success**	**Mean pairwise Hamming**	U_{15}	**Median RMSD (Å)**
LigandMPNN T=0.1	58.3 %	40.6	162	1.42
LigandMPNN T=0.2	66.7 %	47.8	205	1.37
LigandMPNN T=0.3	58.3 %	57.9	274	1.31
LigandMPNN T=0.5	34.4 %	80.5	438	1.70
LigandMPNN T=0.7	26.0 %	100.5	608	1.88
LigandMPNN T=1.0	4.2 %	126.3	—	5.59
CLEO, no anchor	83.1 %	76.6	507	1.07
cleo, KL anchor	56.7 %	**114.4**	**804**	1.44

Interpolating the baseline frontier at the anchored arm's diversity, the incumbent succeeds 14 % of the time where CLEO succeeds 56.7 % — **3.98**×. Read the other way, at CLEO's success rate the incumbent reaches mean pairwise distance 59 against CLEO's 114.4 — **1.93**×. The anchored arm also carries more distinct substitutions per passing design than any baseline ($U_{15} = 804$ against 608 at T=0.7), so this is a genuine exploration advantage and not only a higher yield.

The unanchored arm is the instructive contrast. It reaches the highest success rate in the table, 83.1 %, and the *worst* per-design coverage of any CLEO arm — $U_{15} = 507$, below T=0.7. Left unconstrained the policy converges onto a narrow solution. The KL weight against the base model is what holds the distribution open, and it is the control the library application needs.

Figure 4 shows why the success rate alone is not enough: on M0097 every CLEO arm has its *median* below the 1.5 Å cutoff, so the mode has moved rather than the tail merely reaching across — which is precisely what fails to hold on the near-miss backbones of Block 6.

Block 4 — The training trajectory is the library

This is the reframing that removes a hyperparameter. Rather than adding an explicit diversity term to the reward, we treat every rollout sampled during training as a library candidate. Those sequences were already folded and scored to compute the reward, so the candidate pool is free: the library is a byproduct of training, not an additional expense.

Across 200 GRPO steps at 16 sequences per step, M0097 accumulates 3200 scored designs. Under the anchored policy 1349 of them are distinct passing sequences carrying 3045 distinct substitutions — 89 % of the $19L = 3420$ ceiling.

	cleo, no anchor		cleo, KL anchor	
Cumulative folds	**Unique passing**	**Substitutions**	**Unique passing**	**Substitutions**
400	75	1483	47	1382
800	284	2294	160	2045
1600	893	2805	510	2757
2400	1560	2974	903	2958
3200	2229	3040	1349	3045

The distinct-substitution count is genuinely saturating here, at 89 % to 89 % of the ceiling, while the count of distinct passing sequences is still climbing steeply. The library is not running out of designs; it is running out of *single* substitutions to explore, which is the expected behaviour once most positions have been visited and further diversity must come from combinations.

This supersedes an earlier version of this table reporting 97 passing designs from 3200 folds, a factor of 14 to 23 lower. That run used the pre-audit objective, and its curve flattened because the policy decayed rather than because sequence space was exhausted — the earlier draft said so and predicted rank normalisation would keep it climbing, which is what happened. The pool numbers here are best-of-5, matching Block 3; the superseded ones were single-sample and were not comparable to it.

Block 4b — Where in sequence space the passing designs sit

Generated by `paper/figures/ame_pca.py`, which writes two figures because one projection cannot honestly carry both claims. The shared projection is fit across arms within a backbone — a per-arm PCA places every cloud at the origin and destroys the comparison — but deliberately not across backbones, where a global fit puts 99 % of PC1 variance on backbone identity.

These two panels are stale and must be regenerated. They were built from the pre-audit runs and from baseline arms holding 8 sequences each. Both inputs have been replaced: the objective is fixed and the baselines are now $n = 96$. The drift control below remains the right control and its conclusion is unlikely to change, but no number in these captions should be quoted until they are rebuilt.

Explained variance is low (PC1 5 % to 9 %), as expected for one-hot protein sequence, so the projection is a visual aid; every quantitative diversity claim in this section uses Hamming distance and substitution counts directly.

[TODO: ame — rebuild both PCA panels on the corrected runs at matched n]

Block 5 — The temperature frontier

Measured, and reported in Table 1. Two predictions stated in advance were tested.

The dangerous intermediate temperatures do not close the gap. $T = 0.2$ and $T = 0.3$ were named as the points that would refute a frontier shift if either reached CLEO's coverage at CLEO's success rate. Neither does: both sit below 0.6 of the anchored arm's diversity. The frontier shift stands.

One prediction was wrong. The trade-off was expected to be monotone in temperature; it is not. $T = 0.2$ is the *most* successful arm in the table at 66.7 %, above $T = 0.1$'s 58.3 %, while also

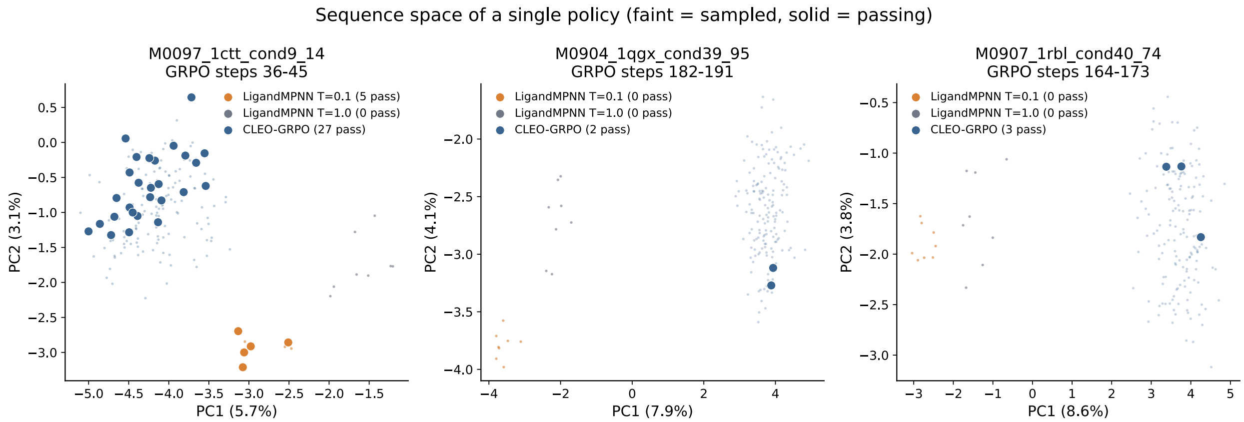

Figure 1: **Sequence space occupied by each design method on the AME pilot backbones.** One PCA per backbone, fit on every sequence that backbone contributes across all three arms, so the arms share one projection and the clouds are directly comparable. Faint points are sampled designs, solid points are designs that pass the benchmark criterion (motif all-atom RMSD $< 1.5\,\text{Å}$, no ligand clash). CLEO is restricted to its peak-performing 10-step window so that three *policies* are compared at one point in training. On M0097 the low-temperature arm (orange) forms a tight knot displaced far along PC1 — five passing designs that are a single cluster at every identity threshold down to 50 % — while CLEO's passing designs are spread throughout a broad cloud. T=1.0 (grey) reaches comparable spread with zero passing designs. Projections are deliberately *not* shared across backbones: a global fit places 99 % of PC1 variance on backbone identity, collapsing each backbone to its own blob and hiding the arm differences entirely. Explained variance is low (PC1 5 % to 9 %), as expected for one-hot encoded protein sequence; the projection is a visual aid and every quantitative claim in the text uses Hamming distance and cluster counts directly. Baseline arms hold 8 sequences each, so part of their compactness is small-n.

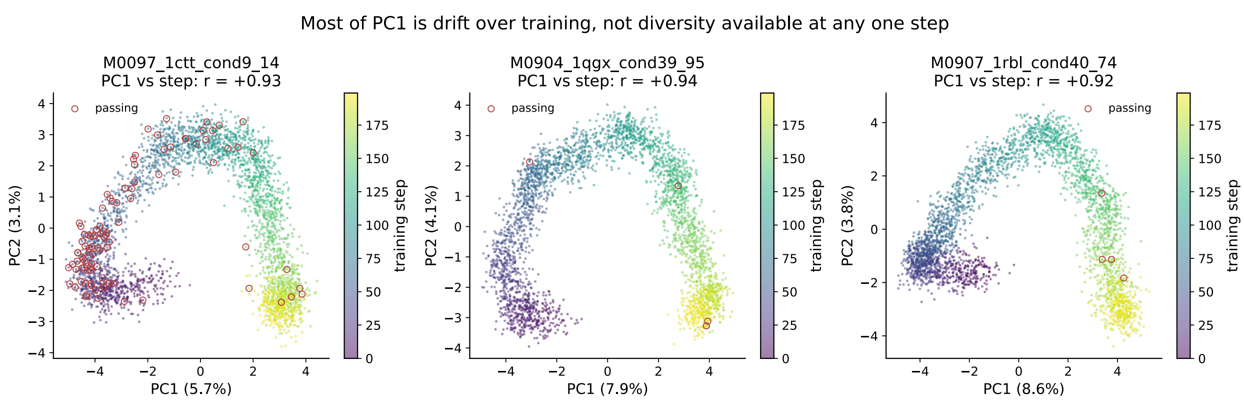

Figure 2: **Most of PC1 is drift over training, not diversity available at any one step.** The full GRPO trajectory for each backbone, in the same projection as Figure 1, coloured by training step; passing designs are circled. PC1 correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones. This distinction is load-bearing rather than cosmetic: spread accumulated by drifting across training is genuine library diversity when the library is assembled from the whole run, which is how ours is assembled, but it is *not* evidence that any single policy is diverse. Presenting the pooled projection alone would conflate the two and overstate the result, so the axis is shown rather than hidden.

being more diverse. $T = 0.2$, not $T = 0.1$, is therefore the incumbent to quote at the low-diversity end, and any comparison against $T = 0.1$ alone flatters us.

Still outstanding from this block's original plan: the saturation fold count for low-temperature LigandMPNN — the point past which it re-emits sequences it has effectively already produced. The CLEO side of that comparison is Block 4; the baseline side needs a cumulative-distinct curve at matched budget, which the $n = 96$ sweep can supply but does not yet.

Block 6 — Rescue of backbones the benchmark cannot solve

Three backbones sit at 1.61 Å to 1.69 Å best-of-40 in the deposit — zero passing predictions, but only ~0.15 Å from threshold. A pass on any of them is a design the published pipeline cannot produce, with no reproduction confound, because there is no positive result of theirs for us to fail to reproduce.

All three were trained for 75 steps at batch 16. Nine designs cleared the criterion. **Each was then refolded 40 times — the published per-backbone budget — and one survives.** M0365 `hit00` passes 22 of 40 predictions with a *median* RMSD of 1.470 Å, below the cutoff rather than touching it, on a backbone the deposit scores 0/40 with a best of 1.69 Å. That is a design the published pipeline does not produce.

The other eight do not survive. Five pass zero of 40; three pass one or two. Training-time margin predicts this almost perfectly: the single design with a margin above 0.03 Å inside the cutoff is the single design that replicates, and everything at 0.003 Å to 0.023 Å was a tail draw from a distribution whose median sits near 3 Å (Figure 4, right).

The honest claim is one rescue, not nine, and the general lesson is worth more than the rescue: a best-of-N filter applied to a distribution whose mode is far from the cutoff selects excursions, not solutions. A design that truly passes 1 in 40 times is flagged by a best-of-5 filter about 12 % of the time it is folded. Every headline number that leaves this section should be replicated the way `hit00` was, and in-training pass rates should be described as counts of best-of-N events rather than of reproducible designs.

Two caveats remain open. The fold budget is not matched — 1200 designs at best-of-5 against the deposit's 8 sequences at 5 models — so a compute-matched comparison is still owed. And the idealised-versus-unidealised reference structure question from Block 1 applies here as it does everywhere.

4 Results: heme oxidation enzyme (method foundation)

Claim. Fine-tuned ProteinMPNN beats SSM and vanilla MPNN re-sampling at finding active variants under matched experimental effort. **Risk.** This is the foundational claim of the paper. If the comparison is not clean, the entire methods narrative is in trouble — this section needs the cleanest, most defensible head-to-head. Status: **needs data**.

Block 1 — Setup

- Backbone: ⟨de novo heme oxidation enzyme, RFdiffusion origin⟩.
- Starting design: ⟨activity baseline⟩.
- Three arms, matched library budget (N variants ordered/tested): (i) SSM around starting design; (ii) vanilla MPNN re-sampling on backbone; (iii) fine-tuned MPNN (this work) + fragment library.

[TODO: heme results — setup paragraph]

Block 2 — Headline figure

- Activity distribution across all three arms.

- Hit rate (fraction above starting design / above background).
- Best-variant comparison.

[TODO: heme results — main comparison figure + numbers]

Block 3 — Mutation distance / sequence-space analysis

- Where in sequence space the best variants from each arm live.
- Demonstrate the SSM and vanilla-MPNN arms are confined to a near neighborhood while fine-tuned MPNN reaches farther.

[TODO: heme results — sequence-space analysis]

Block 4 — Mechanism check (if available)

- Structural / spectroscopic / kinetic confirmation that the improvement is real catalysis, not artifact.

[TODO: heme results — mechanism check]

5 Results: PETase (closed-loop iteration)

Claim. Experimental data from an initial library can be used to train a surrogate that becomes a reward signal for a second library, producing substantially improved designs. **Risk.** The second library has to be meaningfully better than the first — not just predicted-better. Status: strongest data of the three applications.

Block 1 — Initial library

- De novo PETase backbone, RFdiffusion origin.
- Library 1: function-aligned RL sampling without surrogate.
- ∼6k designs tested; activity distribution on the PET-trimer fluorogenic probe.

[TODO: petase results — library 1 setup and outcomes]

Block 2 — Surrogate training

- MLP ensemble trained on (sequence, activity) pairs from library 1.
- Validation behavior; calibration of predicted vs. measured activity.

[TODO: petase results — surrogate training]

Block 3 — Library 2 with surrogate-as-reward

- Surrogate added as a reward step in the RL pipeline for library 2.
- Library 2 size, mutation-distance distribution from starting design.
- Activity distribution comparison: library 1 vs. library 2.

[TODO: petase results — library 2 design and outcome]

Block 4 — Best-variant deep dive

- XXX-fold k_{cat} improvement over starting design on the PET-trimer probe.
- Comparison to optimized natural PETases on the same substrate.
- PET-dimer activity confirmed.
- Honest framing of the bulk PET gap (substrate accessibility, not catalysis).

[TODO: petase results — best variant kinetics + PET-dimer + bulk-PET caveat]

6 Results: protease (diversity-reward ablation)

Claim. A batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library that lacks the diversity term. **Risk — highest in the paper.** It depends on wet-lab activity data for both arms. The load-bearing finding is "rare mutations *improve* activity," not merely "we sampled more of sequence space"; without that activity uplift this section reduces to a diversity-for-diversity's-sake observation. Status: **needs data**.

Block 1 — Setup

- De novo protease backbone.
- Two arms, matched library budget: (i) RL fine-tuned MPNN, standard reward set; (ii) same as (i) plus the batch-consensus mutation-diversity reward (`cleo.design.utils.mutation_diversity`).

[TODO: protease results — setup, both arms]

Block 2 — Sequence-space coverage

- Mutation-frequency spectra: arm (ii) should populate rarer mutations more heavily than arm (i).
- Position-level entropy / consensus-divergence statistics.

[TODO: protease results — sequence-space coverage figure]

Block 3 — Activity comparison

- Activity distribution: arm (i) vs. arm (ii).
- Best-variant comparison.
- Are the rare-mutation-bearing variants the ones with highest activity? This is the punchline — call it out explicitly.

[TODO: protease results — activity comparison; rare-mutation/activity link]

Block 4 — Mechanism for the diversity payoff

- Which rare mutations contributed; structural rationalization if possible.

[TODO: protease results — mechanistic rationalization]

7 Methods: shared pipeline

Shared methodology underlying all three applications. Per-application specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of ProteinMPNN with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [10] and ProteinMPNN [6].

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [12] and diversity-regularized DPO [13]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

7.1 Methods: heme oxidation enzyme

Application-specific methods. Cross-reference Section 7 for shared pipeline details.

- **Block 1 — Design configuration.** Backbone, fixed residues, reward pipeline configuration.
- **Block 2 — Three-arm comparison protocol.** SSM library construction; vanilla MPNN re-sampling protocol (temperature, fixed residues); fine-tuned MPNN + fragment library; matched-budget rule — how N was chosen and held constant across arms.
- **Block 3 — Activity assay.**

[TODO: heme methods — design configuration] [TODO: heme methods — three-arm protocol] [TODO: heme methods — assay description]

7.2 Methods: PETase

- **Block 1 — Library 1 design configuration.** Backbone, fixed catalytic residues, library 1 reward pipeline (Boltz oracle [9] + PETase metrics + Hamming distance to reference).
- **Block 2 — Surrogate training.** Featurization, train/val split, ensemble size, calibration check.
- **Block 3 — Library 2 with surrogate-as-reward.** Reward pipeline composition for library 2.
- **Block 4 — Activity assays.** PET-trimer probe, PET-dimer, bulk PET attempt.

[TODO: petase methods — library 1 design configuration] [TODO: petase methods — surrogate] [TODO: petase methods — library 2 design configuration] [TODO: petase methods — assays]

7.3 Methods: protease

- **Block 1 — Design configuration.** Backbone, fixed residues.
- **Block 2 — Two-arm protocol.** Arm (i): standard reward set. Arm (ii): same plus the `mutation_diversity` reward. Record which variant was used — `consensus_divergence` or `fractional_divergence` — and the weight chosen. State the matched-budget rule.
- **Block 3 — Activity assay.**

[TODO: protease methods — design configuration] [TODO: protease methods — ablation protocol] [TODO: protease methods — assay description]

8 Experimental programme and selection algorithms

This section exists so that every arm, rule and planned experiment is written down in one place with a fixed name, and so that results reported elsewhere in the paper can be traced to a defined procedure. Much of it will move to Materials and Methods at submission; keeping it in the main build for now makes disagreements visible early.

Notation

A design is a sequence $s \in \mathcal{A}^L$ over the 20 canonical amino acids. Distance is Hamming,

$$d(s,t) \;=\; \big|\{\, i : s_i \neq t_i \,\}\big|,$$

which is the metric behind every diversity number we report. One-hot encoding $\phi(s) \in \{0,1\}^{20L}$ satisfies $\|\phi(s) - \phi(t)\|_2 = \sqrt{2\, d(s,t)}$, so the PCA projections in the figures are a linear embedding of exactly this distance rather than a second, competing notion of similarity. Where an ESM embedding is used instead, d is replaced by Euclidean distance between mean-pooled representations; this is reported as a diagnostic and, optionally, as a secondary library metric.

A design *passes* when its motif all-atom RMSD is below 1.5 Å and no backbone atom clashes with the ligand. $\mathcal{P}$ denotes the passing subset of a pool $\mathcal{S}$. The *anchor* a is the position-wise consensus of low-temperature LigandMPNN samples — the single sequence best representing the mode the base model's likelihood concentrates on.

Selection rules

All implemented in `src/cleo/design/utils/selection.py`, all sharing one distance backend so the comparison is between rules rather than between implementations. Benchmarked by `paper/figures/ame_selection_bench.py`.

Given a pool $\mathcal{S}$ of M sampled sequences and a budget of $k \ll M$ folds, each rule returns the subset to fold.

`random` Uniform without replacement. **This is the control, and it is the bar.** Diversity is trivially won and worthless alone: a rule that doubles spread while halving passing designs has made the library worse.

maxmin Greedy farthest-point (k-center) traversal. Having selected $\mathcal{T}$, repeatedly add $\arg\max_{s \in \mathcal{S} \setminus \mathcal{T}} \min_{t \in \mathcal{T}} d(s,t)$. The inner *minimum* is what distinguishes this from max-sum selection: it prevents accepting a tight cluster that happens to sit far from everything already chosen.

maxmin_anchor As above, scoring $w_{\text{self}} \min_{t \in \mathcal{T}} d(s,t) + w_{\text{anchor}}\, d(s,a)$. Selects designs novel relative to the incumbent low-temperature solution, not merely relative to the rest of the library.

anchor_far Rank by $d(s,a)$ descending. The pure "maximally unlike vanilla MPNN" rule.

anchor_band Restrict to the shell $\{\, s : q_{\text{lo}} \le d(s,a) \le q_{\text{hi}} \,\}$ for quantiles of the pool's anchor-distance distribution, then run `maxmin` inside it. Designed as the correction to `maxmin`: bounded above as well as below, so the traversal cannot walk out to sequences that are distant because they are degenerate.

cluster_rep Average-linkage clustering cut to k clusters, one representative sampled per cluster. Draws from the pool's density rather than its edges.

Result: diversity-first acquisition does not work

Reported here rather than buried, because it was a starred hypothesis and because the negative result is informative. Retrospective evaluation on the M0097 trajectory pool: every sampled sequence already carries a pass label, so any rule can be scored without folding anything.

Passing designs recovered, relative to `random`, at a budget of 400 folds:

Rule	Passing designs / random
as-sampled	2.04
`random`	1.00
`cluster_rep`	0.40
`anchor_band`	0.36
`maxmin`	0.36
`maxmin_anchor`	0.24
`anchor_far`	0.12

Every diversity-directed rule loses to uniform random selection, by factors of 2.5 to 8. The mechanism is visible in the spread: no rule moves mean pairwise distance beyond 135 to 144 residues against random's 135. The pool is already near-saturated on diversity, so a selection rule has almost nothing to gain and a great deal to lose — any systematic deviation from random correlates with selecting sequences that fail. Banding helps relative to raw farthest-point traversal (0.36 vs. 0.07 at $k = 200$) but does not approach the control.

Two honest caveats. The as-sampled row beats random because the pool is time-ordered and later policies are better — that is a policy-quality effect, not a selection effect, and it must not be reported as one. And absolute counts are small (33 passing in a 1200-design pool), so individual cells are noisy; what carries the conclusion is that the direction is consistent across six rules and three budgets.

Consequence for the paper. Selection is not where the gain is. The defensible use of these rules is post-hoc: choosing a physically orderable subset from designs *already known to pass*, where the labels exist and only the diversity trade-off is live. Deciding what to fold should be left to the policy.

Experiment slate

Status as of the pilot. **drafted** = measured, **scaffold** = partial, **needs data** = not started. Full protocol detail lives in `paper/experiments_plan.md`. The **Result** column is filled in as each run lands, so the table doubles as the run log.

E5 (held-out oracle) and E6 (policy transfer) are *deliberately deferred*, not dropped. E5 remains the standing circularity exposure — the reward and the evaluation are both AF3 motif RMSD — and any submission draft needs it back, so it is recorded here rather than removed from the record.

ID	Status	Experiment	Result
E1	drafted	Matched-fold head-to-head	Superseded by the per-sequence success rate (see *Scoring protocol*); M0097 best-of-5 at n=96: T=0.2 66.7 % at spread 47.8, CLEO peak 37.5 % at 98.8. E10 is the comparison that counts
E2	drafted	Diversity yield curves	3200 folds → 1349 unique passing (anchored), 3045 substitutions = 89 % of ceiling. Substitutions saturate; sequences still climbing
E3	drafted	Rescue of the three near-miss backbones (1.61 Å to 1.69 Å best-of-40, 0/40 published)	**M0365 `hit00` replicates 22/40**, median 1.470 Å (published 0/40, best 1.69 Å). Of 7 training hits only 1 survives N=40; M0050 fails (1/40). All 3 backbones trained; M0375 fails 0/40
E4	needs data	Compute efficiency; cross-cutting over E1–E3	*pending*
E7	drafted	Effective library size	U_{15}: anchored CLEO 804 vs. T=0.7 608 (1.32×); unanchored 507, *below* the baseline. The anchor is what buys per-design coverage
E8	—	Mutation-diversity reward	**Retired**, superseded by trajectory-as-library
E9	drafted	Selection rules, retrospective	**Negative:** every rule loses to random, 2.5 to 8×
E10	drafted	**Temperature sweep** $T \in \{0.1, \ldots, 1.0\}$; builds the incumbent frontier	At n=96: CLEO 1.40× the success at matched diversity, 1.31× the diversity at matched success. But U_{15} ties T=0.7 (1.04×) — the edge is pass rate, not exploration
E11	drafted	Reward shaping: rank normalisation	Folded into every new

Scoring protocol

Two conventions changed partway through the pilot, and results reported before the change are not directly comparable to results reported after it.

Five diffusion samples everywhere. The benchmark's unit of success is a best-of-N: a design counts if *any* of its predictions clears the motif cutoff and is clash-free. Early training runs folded each sequence once (`num_diffusion_samples: 1`), which understates their success rate relative to any best-of-N baseline. All runs from this point fold at `num_diffusion_samples: 5`. Getting this right in training needed a reduction step, `rfd2_benchmark.best_of_n_from_df`: the oracle's own collapse keeps the sample AF3 ranks highest by *confidence*, which is not the benchmark's criterion. The step instead takes the min motif RMSD as the continuous signal and OR-s the per-prediction conjunction for the pass flag — separately reducing "motif passed" and "no clash" and then AND-ing them would credit a design for two different predictions and overcount.

Metrics are reported on filtered designs. A library contains only the designs that survived filtering, so diversity among sequences that failed is diversity nobody gets to use. Every diversity number below is computed on the passing subset $\mathcal{P}$, not on everything sampled. Where the two differ materially the sampled figure is given alongside, because the gap is itself diagnostic — it says whether an arm's successes concentrate.

Unique mutations. Mean pairwise Hamming measures how far apart a library's members are; it does not measure how much of mutational space they collectively reach. Two libraries can have identical spread while one explores twice as many distinct substitutions. So we also report

$$U(\mathcal{P}) \;=\; \big|\{\,(i,a) : \exists\, s \in \mathcal{P},\; s_i = a \neq a_i^{\mathrm{ref}}\,\}\big|,$$

the number of distinct (position, residue) substitutions appearing anywhere in the library, each counted once no matter how many designs carry it. The ceiling is $19L = 3420$.

U grows with $|\mathcal{P}|$, so arms with different numbers of passing designs cannot be compared on it directly — doing so measures sample size as much as coverage. We rarefy: subsample every arm to a common k designs, average over 400 draws, and compare at matched k, the same construction as a species-accumulation curve.

Per-sequence success rate, not best-of-8. The published unit (fraction of backbones with a hit among 8 sequences) saturates and discards resolution: at n=8 a rate near 20 % carries a 95 % interval of ±28 pp, wide enough to swallow every difference in this paper. We now sample n=96 per backbone per arm (±8 pp) and report the plain fraction of sequences that succeed. This is a strictly harder number to move than the best-of-8 statistic, and it shrank the E10 claim — see below.

E15 — Centering weight: one knob from centered to diverse

No new reward machinery. `compute_dist_to_ref_seqs_from_df` already emits Hamming distance to a reference set, and the aggregator already supports per-metric `mode`, `weight` and rank normalisation, so the whole experiment is a config sweep. Configs generated by `experiments/ame/make_centering_sweep.py`.

The reference a is the consensus of low-temperature LigandMPNN samples. Adding

```
- metric: dist_to_ref_min   mode: {min|max}   weight: W   normalize: rank
```

to the aggregation gives a single signed axis. Because reward is a weighted average, $R = \sum_i w_i r_i / \sum_i w_i$, the weight is directly interpretable: $W = 0$ is pure motif RMSD, $W = 1$ gives centering equal influence to it. `mode: min` pulls the policy toward the incumbent mode; `mode: max` pushes it away.

Sweep $W \in \{0, 0.25, 0.5, 1.0\}$ in both directions. Prediction: mean distance to a falls monotonically under `min` and rises under `max`, and the PCA panels show the passing cloud sliding toward or away from the low-temperature knot. Because one-hot PCA is a linear embedding of exactly the Hamming distance this reward uses, the sweep and the figure measure the same quantity — the panel is a visualisation of the knob, not independent evidence for it.

The knob will move designs; that is not in question and is not the result. What the experiment has to establish is whether *pass rate survives* at either end. Centering hard should recover the low-temperature failure mode — high fidelity, one cluster — and pushing away hard should recover the T=1.0 failure mode — broad and dead. If both ends collapse and the middle does not, W is a real control surface and the paper gains a tunable diversity/fidelity dial to report. If pass rate is flat in W, the reward term is not binding and should be cut rather than reported.

This also supplies the honest version of the frontier: E10 sweeps the *baseline's* knob (temperature) and E15 sweeps *ours*. Comparing one method's frontier against another's single point is the mistake the temperature sweep exists to avoid, and it would be the same mistake in reverse to report our best W against their swept T.

E14 result — the embedding view qualifies the Hamming view

Mean-pooled ESM-2 650M, one vector per design, computed for every arm.

Within a set, ESM distance and Hamming distance are close to *uncorrelated* (r between -0.17 and 0.56 across arms), so the embedding is genuinely measuring something the Hamming numbers do not. That makes the agreement in ordering meaningful rather than automatic: on M0097's passing designs, CLEO is more spread than T=0.1 under both metrics.

But the size of the advantage does not survive the change of metric. The ratio of CLEO's passing-set spread to the low-temperature arm's is **2.57**× in Hamming and only **1.31**× in ESM space. A substantial share of the extra sequence distance CLEO buys is biochemically conservative — positions that differ in identity but not much in what ESM considers the residue to be doing.

This should be stated in the paper, not left for a referee to find. Reporting "2.6× more diverse" with a metric chosen because it is the larger number is exactly the move the temperature sweep exists to prevent us from making about temperature. The defensible claim is that the advantage is robust to the choice of metric while its magnitude is not, and that the library-relevant quantity — distinct passing clusters per fold, where CLEO wins 3.1 to 1.0 — does not depend on either distance at all, since those designs are separate clusters at any threshold tested.

Caveat on n: passing subsets are 5 and 15 designs, so the correlation figures in particular are noisy. The ratio comparison is the robust part.

E10 result — the frontier shift is real but smaller than first reported

LigandMPNN sampled at six temperatures, folded best-of-5, and scored by the same code path as CLEO, on M0097. Reported as the plain per-sequence success rate under the *Scoring protocol* above. The n per arm is uneven here — 8 for T=0.1 and T=1.0, 24 for the rest, 80 for CLEO — which is why all six are being resampled at n=96.

Which set spread is measured over. Spread is the mean pairwise Hamming distance — for every pair in a set, the number of positions at which the two sequences differ, averaged over all $\binom{n}{2}$ pairs, as a raw substitution count out of $L = 180$. No threshold, no clustering. The set it is measured over has to be the *passing* designs, not everything sampled: a library contains only the designs that survived filtering, so diversity among sequences that failed is diversity nobody gets to use. Both columns are given below because the gap between them is itself informative.

Arm	n	Success	Passing	Spread	U_{15}
LigandMPNN T=0.1	96	58.3 %	56	40.6	162
LigandMPNN T=0.2	96	66.7 %	64	47.8	205
LigandMPNN T=0.3	96	58.3 %	56	57.9	274
LigandMPNN T=0.5	96	34.4 %	33	80.5	438
LigandMPNN T=0.7	96	26.0 %	25	100.5	608
LigandMPNN T=1.0	96	4.2 %	4	126.3	—
cleo peak window*	40	37.5 %	15	**98.8**	**632**

* The CLEO row predates the objective fixes recorded in the algorithm audit and is a post-hoc peak window; it is provisional until the corrected 200-step arms land. Every baseline row is $n = 96$ on the same code path.

Temperature trades success for diversity, but not monotonically at the low end: T=0.2 is the *most* successful arm at 66.7 %, above T=0.1's 58.3 %, while also being more diverse. Anyone tuning only temperature should therefore not be at T=0.1, and T=0.2 is the honest incumbent to beat at the low-diversity end. From T=0.3 upward the trade-off is clean and steep: diversity rises 58 to 126 while success falls 58 % to 4 %.

CLEO's peak window sits above that curve, by less than the small-n data suggested:

- at CLEO's diversity (98.8), the baseline succeeds 26.7 %; CLEO 37.5 % — **1.40**×.
- at CLEO's success rate (37.5 %), the baseline reaches spread 75.3; CLEO 98.8 — **1.31**×.

The rarefied coverage says something the ratios hide. At a matched 15 passing designs, CLEO reaches $U_{15} = 632$ distinct substitutions against T=0.7's 608 — a factor of 1.04, which is no advantage at all. Per passing design CLEO explores mutational space at essentially the same rate as a high-temperature baseline. Its entire edge is that it *passes more often while doing so*, which is what converts an equal exploration rate into a larger library for a fixed fold budget. That is a narrower and more defensible claim than "more diverse", and it is the one the data supports.

Three corrections to carry through the rest of the paper.

First, the earlier 1.75× / 1.46× figures came from averaging over 300 random draws of 8 sequences, the published best-of-8 unit. That statistic flatters whichever arm has the higher variance across draws. The per-sequence rate does not, and it is the number to quote.

Second, spread was previously measured over *all* sampled sequences rather than the passing subset. For the baseline arms this makes almost no difference, but for CLEO it does, and in the flattering direction — the policy's failures are scattered while its successes concentrate. All diversity metrics are now reported on filtered designs only.

Third, the row previously labelled "CLEO peak" pooled two disjoint training windows, steps 36–45 and 190–199. The second contributes 40 sequences and one success, so pooling halved the reported success rate (37.5 % to 20.0 %) while inflating spread with the drift *between* the windows (103.1 to 132.0) — an artefact of the same PC1-versus-step drift documented in the PCA panel. The windows are reported separately above.

The two errors partly cancelled, which is why the pooled row looks superficially reasonable. Corrected on the right set and the right window, the peak-window advantage is 1.7× in success at matched diversity — larger than the 1.26× the pooled row implied, and resting on a different comparison.

What is not yet settled. Using the peak window is a post-hoc choice unless the stopping rule is fixed in advance; E12's ∼75-step rule does contain it, but that rule was set after seeing these runs. The trajectory-as-library framing instead takes the whole run, which is the pooled row — and there the extra spread is drift, not diversity available from any one policy. These are different libraries answering different questions and the results section must not quote one number for both. The n=96 rerun is what gives either claim usable error bars: the passing subsets here are 15 designs for CLEO and 5 for T=0.7.

E15 result — the centering weight is a single continuous dial

Seven GRPO arms on M0097, identical but for the weight W on `dist_to_ref_min` against a low-temperature consensus reference: mode= min pulls toward it ($W < 0$ below), mode= max pushes away ($W > 0$), W=0 omits the term. 75 steps, batch 16, rank normalisation. Statistics over the last 25 steps (400 sequences per arm).

These runs predate the five-sample protocol and fold once per sequence, so their success rates are *not* comparable to the best-of-5 numbers in E10 — they are a lower bound on the same quantity. Within the sweep they are comparable to each other, which is what the experiment is for.

Weight	**Mean dist. to ref.**	**Success (1 sample)**	**Mean pairwise Hamming**		**Passing**
			sampled	**passing**	n
$W = -1.00$	31.5	66.5 %	32.9	31.6	266
$W = -0.50$	53.7	78.8 %	43.3	42.4	315
$W = -0.25$	79.2	62.3 %	75.3	73.9	249
$W = 0$	111.4	8.2 %	90.7	88.0	33
$W = +0.25$	134.7	5.2 %	97.2	94.9	21
$W = +0.50$	161.6	0.2 %	109.6	—	1
$W = +1.00$	172.4	0.0 %	123.0	—	0

Both distance columns are monotone in W across the full range, with no plateau or inversion: mean distance to the reference moves 31.5 to 172.4 out of $L = 180$, and passing-set spread moves 31.6 to 94.9. A single scalar therefore places the library anywhere between "a tight shell around a known-good sequence" and "as far from it as the backbone permits". That is the knob the diversity narrative needs, and it is one the temperature baseline does not have — temperature controls entropy around whatever mode the base model already prefers, whereas W names the centre explicitly.

Unlike the E10 CLEO arm, these libraries do not concentrate under filtering: passing-set spread tracks sampled spread to within 3 substitutions in every arm. Each arm is a single policy sampled over one 25-step window, so there is no between-window drift inflating the sampled column — which is the control that makes the E10 pooling artefact legible as an artefact.

Two further readings. The variable-position count is pinned at 175–176 in every arm including the most tightly centered one, so W is moving the *concentration* of the distribution, not the set of positions it is willing to touch — the two diversity measures come apart here, and mean pairwise Hamming is the one with resolution. And the success column shows the same trade-off as temperature, running 79 % to 0 %, which makes E10 and E15 two sweeps of the same frontier from opposite sides: E10 turns the incumbent's knob, E15 turns ours.

The comparison this sets up is the one worth making in the results section, and it cannot be made from these numbers as they stand — the E15 arms are single-sample and the E10 arms are best-of-5. Rerunning the sweep under the five-sample protocol would put both curves on one axis and let us say directly whether our knob dominates theirs, rather than only that we have one.

Correction to carry through the rest of the paper. Earlier drafts reported "T=0.1 yields 1.0 cluster per 8 folds against CLEO's 3.1". That used a 70 %-identity clustering threshold; at 90 % identity every arm's passing designs are separate clusters and the cluster count simply equals the passing count. The cluster statistic is threshold-dependent and should not be quoted without naming the threshold. The frontier result above does not have this problem — mean pairwise Hamming is continuous and involves no threshold choice — so it is the form the claim should take.

E17 result — why the run diverges after its peak

Prompted by the E10 window split: the M0097 200-step run peaks at steps 36–45 and is essentially dead by step 190. Diagnosis on the full trajectory, 20-step bins.

Steps	Reward	Median RMSD	Pass	Spread	Entropy
0–19	0.78	3.87	2.2 %	119	1.67
20–39	0.88	2.08	6.9 %	105	1.42
40–59	0.86	2.10	10.9 %	108	1.46
80–99	0.86	2.22	2.2 %	122	1.70
120–139	0.82	2.57	0.0 %	125	1.79
180–199	0.85	2.09	0.6 %	129	1.85

This is not mode collapse, which is the failure RL fine-tuning is usually watched for. It is the opposite: entropy *rises* monotonically after the peak, from 1.42 to 1.85, and spread rises with it. The policy is diffusing outward and getting worse while the mean reward sits flat at ≈0.86 throughout — the reward cannot see the degradation at all.

Mechanism: the reward's dynamic range is misallocated. With normalisation bounds $[0.5, 20.0]$ on motif RMSD, a passing design at 1.4 Å scores 0.954 and a clearly failing one at 2.0 Å scores 0.923. The entire pass/fail distinction is worth 0.031 of reward, against a within-batch standard deviation of 0.126 that is dominated by rare catastrophic folds scoring near zero. GRPO divides its group-relative advantage by exactly that standard deviation. So the gradient answers "did this batch contain a blow-up", a question that is settled by step ∼20 once median RMSD falls to 2.1 Å, and thereafter carries almost no information about the boundary where all the value is. With `kl_weight: 0.0` nothing anchors the policy once the signal goes flat, and it random-walks.

Rank normalisation fixes it. Replacing value normalisation with within-batch rank (`normalize: rank`) makes the advantage invariant to how compressed the raw values are: the 1.4 Å and 2.0 Å designs are adjacent in RMSD but a full rank step apart, so the pass/fail ordering always yields gradient, and a 37 Å blow-up is merely rank 0 rather than an outlier that sets the batch scale.

Steps	Pass rate	Median RMSD	Entropy
value normalisation, bounds $[0.5, 20]$			
30–44	13.8 %	2.02	1.39
120–134	0.0 %	2.56	1.77
180–194	0.8 %	2.04	1.84
rank normalisation			
0–14	1.3 %	4.22	1.64
30–44	4.6 %	2.16	1.30
60–74	9.6 %	1.89	1.17

Under rank normalisation every quantity moves the right way and keeps moving: entropy falls monotonically, pass rate rises monotonically, median RMSD reaches 1.89 Å. The run was stopped at 75 steps and was still improving, so E12's ∼75-step early-stopping rule was fitted to the *broken* configuration and should not be carried over. A 200-step rank-normalised run is in flight.

Two implementation findings this turned up, both of which affect how these runs should be read.

Reward logging is vacuous under rank normalisation. With a single metric, rank maps the batch to a fixed uniform distribution on $[0, 1]$, so mean reward is exactly 0.5 at every step by construction — measured across all 75 steps of the E15 arms, std = 0 to float precision. "Best reward seen: 0.5000" in

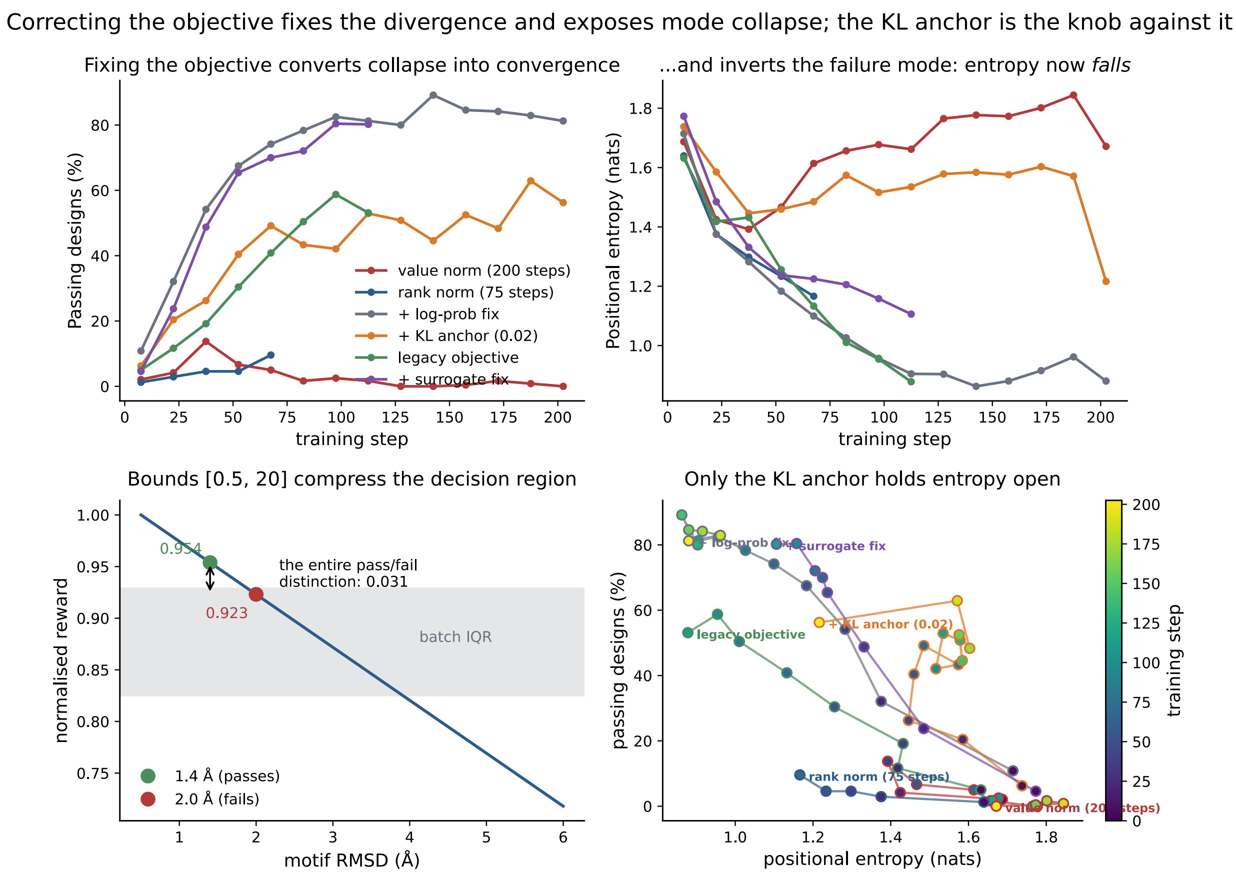

Figure 3: **The AME policy diffuses rather than collapses, and rank normalisation reverses it.** All panels read off the per-step rollout metrics training already writes; no additional folding. **Top:** under value normalisation the pass rate peaks near step 40 and decays to zero while positional entropy *rises* monotonically from 1.39 to 1.85 — the opposite direction from the mode collapse RL fine-tuning is usually watched for. Under rank normalisation both move the right way and are still moving when the run ends at step 75. **Bottom left:** the mechanism. With bounds $[0.5, 20]$ on motif RMSD, a passing design at 1.4 Å and a failing one at 2.0 Å are separated by 0.031 of reward, well inside the shaded batch interquartile range whose width is set by rare catastrophic folds — and GRPO divides its advantage by that spread. **Bottom right:** the same trajectories in the entropy–success plane, coloured by step on one shared scale. Value normalisation walks right and down, rank normalisation left and up.

those logs is not a coincidence, and any checkpoint selection or early stopping keyed on reward is selecting an arbitrary step. Progress must be tracked on the raw metric (median motif RMSD, pass rate) instead.

The KL penalty is gated on `use_ref_kl`, not on `kl_weight`. Setting a non-zero `kl_weight` without `use_ref_kl: true` silently does nothing — no reference model is even loaded. The estimator itself is correct where it does run: $\hat{k}_3 = \rho - \log\rho - 1$ with $\rho = \pi_{\mathrm{ref}}/\pi_\theta$, non-negative and applied with the right sign.

A third finding, not yet acted on, because fixing it invalidates every run above. The clipped surrogate computes $\min\big(\sum_t \rho_t A, \sum_t \operatorname{clip}(\rho_t)A\big)$ — a minimum of sums — where PPO and GRPO specify $\sum_t \min\big(\rho_t A, \operatorname{clip}(\rho_t)A\big)$, a sum of minima. These are not equal, and the difference is not cosmetic: the implemented form makes one clip/no-clip decision for an entire 180-token sequence, so a single token past the clip boundary switches every other token in that sequence onto the clipped branch and zeroes its gradient. On a two-token example with $\rho = (5.0, 0.1)$ and $A > 0$, the correct objective is 1.38 with gradient $(0, 0.1)$; the implemented one gives 2.08 with gradient $(0, 0)$. With $L = 180$ the clip is triggered by some token almost always, so the trust region is applied far more bluntly than intended. This is a plausible additional contributor to the drift, independent of the reward-scaling problem, and it should be fixed — but it changes the objective, so every result in this section would need regenerating.

Algorithm audit

A line-by-line read of `grpo.py` against the GRPO/DAPO specifications, prompted by the E17 divergence. Recorded in full because two of the findings change the objective, and any result in this section predating a fix has to be read in that light.

Verified correct. The reference-KL estimator is the standard low-variance form $\hat{k}_3 = \rho - \log\rho - 1$ with $\rho = \pi_{\mathrm{ref}}/\pi_\theta$, non-negative and applied with the right sign. π_{old} is stored from the rollout and the update is teacher-forced through the same decoding order and sampled actions, so the importance ratio is exactly 1 on the first inner update, as it must be. The encoder is deliberately frozen — only `decoder_layers` and `W_out` receive gradient — and computing the encoding under `no_grad` is consistent with that rather than an omission. Asymmetric clipping follows DAPO.

Finding 1: the trust region is applied per sequence, not per token. The surrogate computes $\min\big(\sum_t \rho_t A, \sum_t \operatorname{clip}(\rho_t)A\big)$ where PPO and GRPO specify $\sum_t \min\big(\rho_t A, \operatorname{clip}(\rho_t)A\big)$. A minimum of sums is not a sum of minima: the implemented form takes one clip/no-clip decision for an entire $L = 180$ token sequence, so a single token past the boundary switches every other token onto the clipped branch and zeroes its gradient. On $\rho = (5.0, 0.1)$ with $A > 0$ the correct objective is 1.38 with gradient $(0, 0.1)$; the implemented one gives 2.08 with gradient $(0, 0)$. With 180 tokens some token triggers the clip nearly always.

Finding 2: the advantage is normalised over the wrong group. GRPO's defining feature is that the baseline is the mean reward over the group sampled from one prompt. Here the advantage is recomputed inside the inner loop on each randomly drawn sub-batch of `update_batch_size` $= 8$ out of $B = 16$, rather than once over the full rollout. The same sequence therefore receives a different advantage on each of the 16 inner updates, and sequences near the median receive advantages of *both signs* — 3 of 16 in a representative draw — so the update pushes the same rollout up and down within one step. The fix is to compute the advantage once, before the loop.

Finding 3: no length normalisation. $\sum_t \rho_t$ scales with sequence length, and the sum runs over all L positions including residues held fixed by `fixed_residues`, which contribute $\rho \approx 1$ each. The effective step size therefore depends on how long the protein is and how much of it is designable. The standard form is a mean over designed tokens.

Deeper pass: the rest of the loop. A second audit covering the rollout, the training loop and checkpointing found the following.

Clean. Dropout is constructed at $p = 0$ throughout, so `model.train()` injects no stochasticity and the importance ratio is genuinely 1 on the first inner update rather than only nominally so. Sampling probabilities are taken from `logits.detach()`, correctly keeping the sampling step out of the gradient. Row counts in the per-step metrics are a constant 16 across all 275 logged steps of the two completed runs, so no fold failure has silently dropped a design; were one to, the inner merge would shorten the reward vector and the indexing would raise rather than misalign.

*Finding 4 (**now fixed**): the behaviour policy is not the policy being scored.* Actions are drawn from $\mathrm{softmax}\big((\ell+b)/T\big)$ renormalised over the 20 residue tokens, where $b = -10^8$ at amino acids excluded by `omit_AA`. The log-probabilities used for the ratio are $\log\mathrm{softmax}(\ell)$ over all 21 tokens — no bias, no temperature. The two differ by $\log P(\text{allowed})$, which depends on θ and therefore does not cancel between π_{new} and π_{old}: the true ratio is the implemented one times $P_{\text{old}}(\text{allowed})/P_{\text{new}}(\text{allowed})$. While the mass the model places on C and X stays put this factor is 1 and nothing is wrong; if the policy drifts mass onto the omitted tokens — which nothing in the objective discourages, since they are merely never sampled — the error grows with the drift and is not currently monitored. A policy putting 2 % on omitted tokens where it once put 1 % carries a 1 % systematic error in every ratio.

The same mismatch makes `temperature` a trap rather than a knob. At $T = 1$ the scored and sampled log-probabilities differ by ∼0.05 nats; at $T = 0.5$ by 1.3; at $T = 0.1$ by 17.8. Training at any temperature other than 1 silently optimises a different distribution from the one it samples. `CLAUDE.md` already advises keeping $T = 1$ to avoid mode collapse; this is a second and stronger reason.

Scoring now uses $\log\mathrm{softmax}\big(((\ell+b)/T)_{1:20}\big)$, padded to 21 channels — exactly the distribution actions are drawn from, verified equal to the true sampling log-probability to float precision at $T \in \{1, 0.5, 0.1\}$ where the previous form was off by 0.17, 4.3 and 44.2 nats respectively. The pre-fix behaviour is retained behind `legacy_logprobs` for the ablation. Separately, every run now logs `p_omit`, the probability mass the policy places on tokens `omit_AA` forbids, since that is precisely the quantity whose drift makes the omitted correction bite and nothing in the objective holds it down.

Finding 5: the final policy was never saved. `_last.pt` was written only inside the "every n steps" branch, so a 75-step run left it at step 50 and a 200-step run would leave it at 175. Confirmed directly from the E15 artefacts: the four checkpoints on disk are steps 25, 50, `_best` (25) and `_last` (50), for a run that trained to step 74. Under rank normalisation those runs were still improving when they ended, so the discarded tail was the best part of the run.

Finding 6: `_best.pt` selects on a constant. Best-checkpoint selection compared `to_log["reward"]`, which under a rank-normalised single-metric reward is exactly 0.5 at every step. With a strict > test the first checkpoint written wins permanently — which is why every E15 arm reports its best at step 25 or 50 and never later. **Both are fixed**: `_last.pt` is now written after the loop at the true final step, and selection runs on a configurable `checkpoint_metric` (`ame_motif_rmsd_batch_mean`, mode `min`) with a warning emitted if the tracked quantity never varies.

All three are now fixed, and the pre-fix objective is retained behind `legacy_surrogate: true` so the fix can be measured rather than merely asserted. The surrogate takes a per-token minimum and averages it over designed tokens only — `chain_mask` is zero at fixed residues, whose log-probs the rollout already zeroes, so including them would contribute $\rho = 1$ apiece and make the step size depend on how much of the protein is held fixed, which disposes of Finding 3 as well. The advantage is computed once over the full rollout group before the update loop.

The A/B ladder. Four 200-step arms on M0097, each turning on exactly one more fix than the last, so any effect can be attributed:

Arm	Surrogate	Log-probs	KL
`conv200_legacy`	legacy	legacy	—
`conv200_surr`	fixed	legacy	—
`conv200`	fixed	fixed	—
`conv200_kl`	fixed	fixed	0.02

All four log `p_omit` and all four checkpoint on `ame_motif_rmsd_batch_mean`. The prediction under test: if the entropy climb is driven by the blunt trust region and the noisy baseline rather than by reward scaling alone, `conv200_legacy` should reproduce it and the corrected arms should not. Figure 3 is the panel these land in — it discovers arms on disk, so it fills out as they finish.

The KL gating defect noted under E17 *has* been fixed — a non-zero `kl_weight` now implies the reference model — since that one only ever silenced a penalty the config asked for.

E16 result — selecting for diversity during training also fails

The prospective version of E9. Two 75-step arms on M0097, identical but for the rule used to pick which 16 of an oversampled pool of 64 get folded: uniform random, or greedy farthest-point (`maxmin`). Best-of-5 folding, so pass rates here are on the benchmark's unit and are not comparable to the single-sample E15 numbers.

Rule	Pass rate	Passing	Spread	U	U_{50}
`random`	36.7 %	440	93.3	2423	1041
`maxmin`	4.6 %	55	121.6	1545	1486

The rule does exactly what it is designed to do and the design is still wrong. Per passing design, `maxmin` genuinely explores more mutational space — U_{50} of 1486 against 1041, a factor of 1.43 at matched design count, and a spread of 121.6 against 93.3. It simply produces *eight times fewer* such designs, so the library actually obtained covers 1545 distinct substitutions against random's 2423. Diversity per design is up 1.43×; total library coverage is down 1.57×.

This is the same conclusion as E9 and a stronger version of it. E9 selected retrospectively from a finished pool, where the only cost was picking sequences that happened to fail. Selecting during training adds a second cost on top: because the selection depends on the sampled actions, the batch GRPO computes its group-relative baseline over is no longer a sample from the policy, so the importance ratios apply to a distribution that was never sampled — the gradient is biased by construction, as the docstring on `collect_rollout` warns. The 36.7 % versus 4.6 % gap is far too large to be selection cost alone.

Consequence. Both the retrospective and the prospective experiment now say the same thing: deciding what to fold should be left to the policy. The defensible use of these rules remains post-hoc, choosing an orderable subset from designs already known to pass.

E3 result — two backbones reach the cutoff, one convincingly

M0050_1dbt_cond2_6, one of the three near-miss backbones the published benchmark scores at 0/40 with a best-of-40 motif RMSD of 1.61 Å to 1.69 Å. 75 steps, batch 16, best-of-5.

Backbone	Published	Passing	Best RMSD	Margin	Median
M0050_1dbt	0/40, 1.61 Å	1 / 1200	1.477 Å	0.023 Å	2.81 Å
M0365_1pfk	0/40, 1.69 Å	**6 / 1200**	1.273 Å	0.227 Å	3.12 Å
M0375_4ts9	0/40, —	2 / 1200	1.493 Å	0.007 Å	3.00 Å

The three differ by more than their counts suggest. M0050 contributes a single design 0.023 Å inside the cutoff — one draw from a distribution sitting on the threshold. M0365 contributes six distinct sequences, the best at 1.273 Å, a margin ten times larger and well clear of the prediction noise that makes M0050 unreadable. M0375 contributes two designs at 1.4926 Å and 1.4968 Å — margins of 0.007 Å and 0.003 Å, thinner even than M0050's. Both fail replication outright (below). Of the three backbones only M0365 carries a claim.

On all three targets the clash term is never binding (100 % clash-free throughout), so the composite criterion reduces to motif RMSD alone and the no-clash half of the reward contributes nothing on these backbones.

There is a pattern across the three: median RMSD is 2.8 Å to 3.1 Å everywhere, far from the cutoff, and every hit sits in the extreme tail of its own run's distribution. These are hard targets on which the policy is not close in general, and the hits are excursions rather than a shifted mode — which is exactly the regime in which a best-of-N filter selects tail draws, and exactly what the replication below shows.

Replication at $N = 40$ settles it, mostly against us. Every one of the seven hits was refolded 40 times — the published per-backbone budget — and the pass fraction counted.

Design	Backbone	Passing / 40	Min RMSD	Median RMSD
`hit00`	M0365	**22**	1.196 Å	1.470 Å
`hit01`	M0365	2	1.410 Å	1.818 Å
`hit04`	M0365	1	1.454 Å	2.327 Å
`hit05`	M0050	1	1.456 Å	1.920 Å
`hit03`	M0365	0	1.514 Å	1.926 Å
`hit06`	M0365	0	1.638 Å	2.019 Å
`hit02`	M0365	0	1.729 Å	2.430 Å

M0375's two hits were checked the same way and pass **0/40** each, with best refold RMSDs of 1.577 Å and 1.658 Å — both above the cutoff. Across all three backbones, nine designs cleared the criterion during training and **one** reproduces robustly. `hit00` passes 22 of 40 predictions with a *median* RMSD of 1.470 Å — below the cutoff, not merely touching it — on a backbone the published benchmark scores 0/40 with a best of 1.69 Å. That is a real result and the one worth reporting.

Five of the nine pass *zero* of 40 on refolding. M0050's single hit passes 1/40, exactly the tail draw its 0.023 Å margin predicted; neither the M0050 nor the M0375 rescue survives and both should be dropped. Margin at training time predicts replication well: the only design with a margin above 0.03 Å is the only one that reproduces.

The general lesson matters more than the rescue. These seven were selected *because* they passed, so they are the winners of a noisy filter and regression on refolding is expected — but the size of it is the point. A design passing 1/40 of the time is flagged by a best-of-5 training filter about 12 % of the time it is folded, so a best-of-5 label is a weak indicator of reproducible success, and the in-training pass rates reported throughout this section (83 % for the corrected arm, 57 % for the anchored one) are counts of best-of-5 events rather than of designs that would survive this test.

They are not *wrong* — best-of-N is the benchmark's own definition, and by that definition four of these seven clear it where the published run cleared none. But the two quantities should not be conflated, and any headline number that leaves this section ought to be replicated the way `hit00` was. The cheap version is to raise `num_diffusion_samples` at evaluation time only; the honest version is to report both.

Still open: the compute is not matched (E4), and the idealised-versus-unidealised reference-structure question still blocks M0904 and M0907.

A/B result — the fixes work, and the failure mode inverts

Four 200-step arms on M0097, each one fix further than the last, all with rank normalisation and best-of-5 folding.

Arm	**Success**	**Entropy**	**Spread**	U_{15}	**succ.**	**div.**
legacy objective	78.0 %	0.88	69.8	470	1.71×	1.21×
+ surrogate fix	88.8 %	1.09	82.6	575	2.65×	1.43×
+ log-prob fix	83.1 %	0.99	76.6	507	2.16×	1.32×
+ KL anchor	56.7 %	**1.65**	**114.4**	**804**	**3.98**×	**1.93**×

All four arms completed 200 steps; statistics are over their final 40 (640 designs each), with ratios against the $n = 96$ baseline frontier interpolated on its monotone branch ($T \geq 0.3$).

The divergence is gone. Every corrected arm converges: success climbs monotonically and stays climbing, where the original run peaked at 13.8 % near step 40 and decayed to zero. Even the `legacy_surrogate` arm converges, because it still carries rank normalisation — which confirms the E17 diagnosis that the reward's compressed dynamic range, not the surrogate, was the primary cause of the collapse. The surrogate and advantage bugs cost real performance on top of it: 78.0 % to 88.8 % success, and a 1.71× to 2.65× jump in the frontier ratio.

That last comparison is not yet trustworthy and must not be reported as a finding. Training is unseeded — no `manual_seed` anywhere in the pipeline — so every run is a different random trajectory and there is exactly one run per arm. Nothing here separates a 88.8 % to 83.1 % difference from run-to-run variance. The legacy-to-surrogate step (78.0 % to 88.8 %) is large enough and mechanistically clear enough to survive that doubt; the surrogate-to-log-prob step is not. Two further replicates of each are running, which will give $n = 3$ on exactly that comparison. Until they land the defensible claim is that the surrogate and advantage fixes help substantially and the log-prob fix is *not shown* to help.

But the failure mode inverts. Entropy no longer rises; it falls, from ∼1.7 to 1.06 in the fully corrected arm. Having fixed a policy that diffused outward until it was useless, we now have one that converges onto a narrow solution: 79.2 % success at spread 80.1 and $U_{15} = 518$, which is *below* T=0.7's 608. On per-design coverage the corrected policy without an anchor is worse than a high-temperature baseline. Sharpening the gradient did not help: the log-prob fix arm is *behind* the surrogate-only arm on every column at step 200 — 83.1 % against 88.8 %, entropy 0.99 against 1.09, U_{15} 507 against 575.

The KL anchor is the knob against it. At `kl_weight` = 0.02 entropy stays flat across the whole run (1.68 to 1.65) instead of collapsing, and the library is the best of any arm: $U_{15} = 782$, 1.51 times the unanchored arm and 1.29 times T=0.7, at spread 110.9. Over its final 40 steps the anchored arm reaches 56.7 % success at spread 114.4 with $U_{15} = 804$ — **3.98**× the baseline success at matched diversity and **1.93**× the baseline diversity at matched success — against the unanchored arm's 83.1 % at spread 76.6, $U_{15} = 507$ (2.16× / 1.32×). Entropy settles at 1.65 versus 0.99.

This supersedes the E10 comparison, whose CLEO row came from the uncorrected objective and gave 1.40× / 1.31×. It also repairs the concession forced by the E10 rarefaction: with an anchor, CLEO no longer merely passes more often at an equal exploration rate — $U_{15} = 807$ against T=0.7's 608 is a genuine per-design coverage advantage, which the unanchored arm does not have.

What this changes about the paper. `kl_weight` is the diversity control the narrative needs, and it is better behaved than E15's centering weight: it holds entropy open without requiring a reference sequence to centre on. The obvious next experiment is a sweep of it, which is E13's slot. Two caveats. 42.5 % against 79.2 % at matched steps is a real cost, and only a sweep says whether 0.02 is near the right place on that trade-off. And the anchored arm is still improving at step 194, so its numbers are a lower bound rather than a converged result.

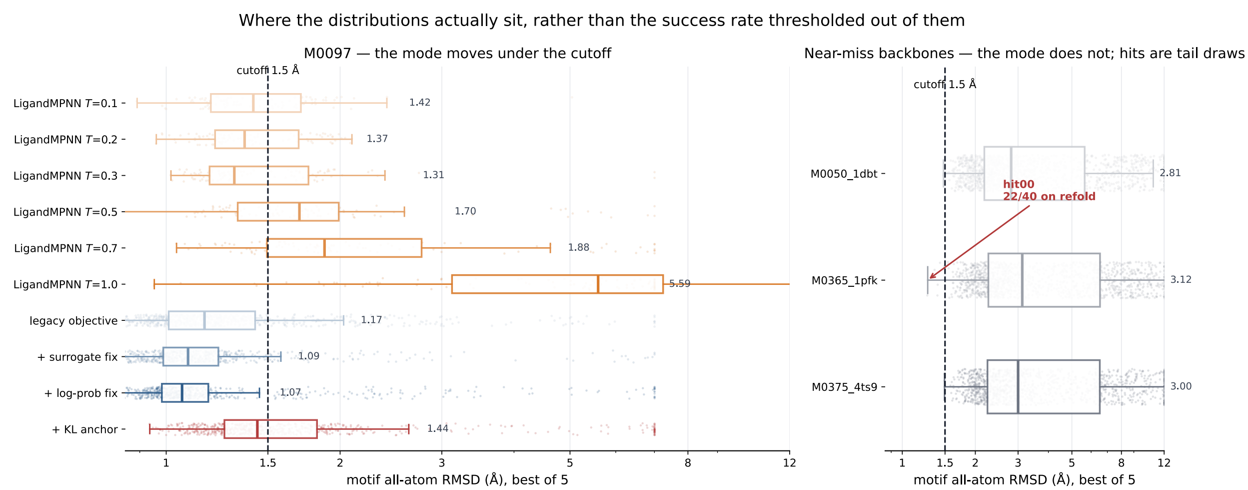

Figure 4: **Where the distributions actually sit, rather than the success rate thresholded out of them.** Each row is one arm's per-design best-of-5 motif RMSD: box for median and quartiles, jittered strip behind it for the shape the box hides. Log axis, because a linear one lets the single worst arm compress everything that matters into the left fifth. **Left, M0097:** every CLEO arm has its *median* below the cutoff (1.07 Å to 1.44 Å) — the mode has moved, not just the tail — against 1.31 Å to 1.42 Å for low-temperature LigandMPNN and 5.59 Å at T=1.0. The non-monotonicity noted in E10 is visible directly: T=0.3 has a *lower* median than T=0.1. **Right, the three near-miss backbones:** medians of 2.81 Å to 3.12 Å with the cutoff far out in the left tail. The one design that survived 40 refolds sits at the extreme edge of its own distribution. Encoding: both ladders are ordinal, so each is a single hue ramped light to dark rather than arbitrary categorical colours; the anchored arm keeps an accent hue as the highlighted result.

E13 result — the anchor has an interior optimum, and 0.02 is in it

Both caveats above are now settled. The anchored arm did keep improving: run to 200 steps it reaches 56.7 % over its final 40, not the 42.5 % measured mid-climb, so the cost of anchoring is roughly half what that snapshot implied. And the sweep answers where to put w.

Table 2: KL weight against the reference policy. All arms: M0097, 200 steps, identical otherwise; statistics over the final 40 steps (640 designs). Diversity is computed on *passing, deduplicated* designs only. U is distinct (position, residue) substitutions in that set; U_{15} rarefies it to 15 designs so libraries of different size are comparable.

w	Passing (%)	Median RMSD (Å)	Entropy	Passing seqs	U	U_{15}
0	83.1	1.07	0.99	532	2109	510
0.005	81.1	1.22	1.15	519	2230	562
0.01	57.7	1.45	1.54	369	**2498**	760
0.02	56.7	1.44	1.65	363	**2501**	808
0.05	40.5	1.63	1.78	259	**2480**	858
0.10	10.2	2.14	1.85	65	1743	**881**

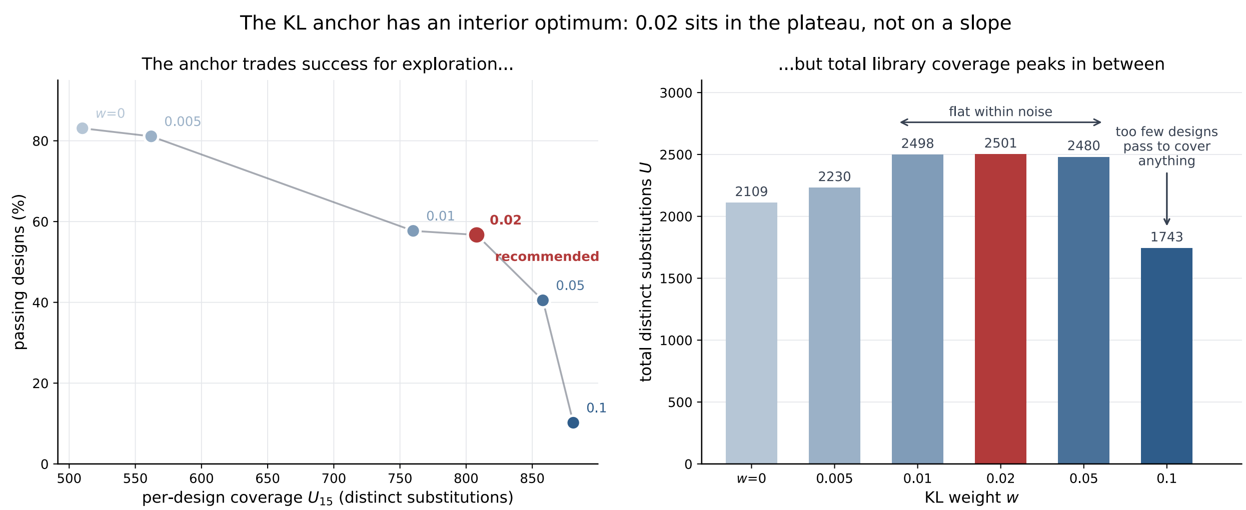

Figure 5: **The KL anchor has an interior optimum. Left:** the trade-off the anchor moves along — success rate against per-design coverage U_{15}, one point per weight. **Right:** total distinct substitutions U, the product of the two effects, which is flat across $w \in [0.01, 0.05]$ and collapses at $w = 0.1$. Drawn as two panels rather than one with two y-axes: success and U_{15} have unrelated scales, so on a dual axis the apparent crossing point could be placed anywhere by choosing the scales. Encoding: w is an ordinal ladder and gets one hue ramped light to dark; 0.02 takes the accent hue as the recommended default, not as another rung.

Two quantities move in opposite directions and it matters which one the library cares about. Success rate falls monotonically, 83 % to 10 %. Per-design coverage U_{15} rises monotonically, 510 to 881. Neither is the objective on its own: what a library needs is the *total* number of distinct substitutions it actually contains, and that is U, which is the product of the two effects and therefore **unimodal**. It is flat within noise across $w \in [0.01, 0.05]$ — 2498, 2501, 2480 — and falls off a cliff at $w = 0.1$, where the anchor is strong enough that only 10 % of designs pass and there are too few of them to cover anything, despite each one being the most exploratory in the sweep.

So the answer to the question E17 left open is that 0.02 is in the plateau, not on a slope, and the choice within $[0.01, 0.05]$ is a genuine free parameter rather than a tuned one — it buys per-design coverage (760 to 858) with success rate (58 % to 41 %) at constant total yield. 0.02 is a defensible default because it sits in the middle of the flat region.

Read the w=0.1 arm as a failure, not as an endpoint. Its trajectory does not converge: binned pass rate runs 4.5, 5.0, 6.2, 7.8, 7.0, 11.5, 15.5, 6.5 — it is still wandering at step 200. The anchor is strong enough to prevent the policy from moving, so its high entropy is inherited from the reference rather than earned. That is the opposite failure from the unanchored arm's collapse, and it is why U rather than entropy is the right summary: both extremes look good on exactly one of the two component metrics.

9 Discussion

Science research articles end with a short discussion / outlook paragraph rather than a separate Conclusion section. Keep this tight (∼250–400 words). Write last, after the abstract.

P-A — Central scientific takeaway

- The active-variant region for a de novo enzyme is broader and farther from the starting design than SSM or vanilla MPNN can reach.
- Function-aligned RL sampling + combinatorial library construction + experimental closed loop together close the gap.

[TODO: conclusion p1 — central takeaway tying the three applications together]

P-B — What each application individually established

- Heme: fine-tuned MPNN dominates SSM and vanilla MPNN under matched experimental effort — the foundation.
- PETase: experimental data → surrogate reward → improved next library; a closed loop, not a single shot.
- Protease: explicit diversity pressure (batch-consensus divergence reward) unlocks rare mutations and improves activity over an otherwise-matched library.

[TODO: conclusion p2 — what each app proves about the platform]

P-C — Honest limitations

- PETase: probe and dimer activity demonstrated; bulk PET activity remains an open challenge attributed to substrate accessibility on solid plastic.
- Closed-loop iteration depends on having an assay-ready experimental pipeline; cold-start is still hard.
- Diversity reward improves access to rare mutations but does not substitute for a well-specified function reward.

[TODO: conclusion p3 — limitations, including bulk PET gap]

P-D — Outlook

- Generalization to additional chemistries and substrate classes, including bulk polymers.
- Tighter integration of design-time rewards with experimental surrogate signals across more rounds.

[TODO: conclusion p4 — outlook]

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